

Stem-and-Leaf Plots

The *stem-and-leaf plot* is a tool of *exploratory data analysis* (EDA) that seeks to reveal essential data features in a simple way. A stem-and-leaf plot is basically a frequency tally, except that we use digits instead of tally marks. For two-digit or three-digit integer data, the *stem* is the tens digit of the data, and the *leaf* is the ones digit.

Example: Exam Scores

Here is a dot plot for exam scores for 22 students, arranged from lowest to highest:

52, 56, 60, 64, 66, 69, 71, 73, 73, 76, 77, 79, 80, 80, 81, 82, 82, 85, 85, 87, 91, 96

The stem-and-leaf plot is:

Frequency	Stem	Leaf	
2	5	2 6	↔ data values 52, 56
4	6	0 4 6 9	↔ data values 60, 64, 66, 69
6	7	1 3 3 6 7 9	↔ data values 71, 73, 73, 76, 77, 79
8	8	0 0 1 2 2 5 5 7	↔ data values 80, 80, 81, 82, 82, 85, 85, 87
2	9	1 6	↔ data values 91, 96
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A stem-and-leaf is like a histogram, in that it uses equally spaced classes and can reveal modality as well as the range. For example, this stem-and-leaf plot shows that most exam scores were in the 80s. The leaf digits have been sorted, although this is not necessary. The stem-and-leaf has the advantage that we can retrieve the raw data, by concatenating any *stem digit* with each of its *leaf digits*.

Example: Drug Prices

The U.S. government's Medicare Plan Finder allows consumers to compare costs of Part D prescription drug plans offered by retail pharmacies. For zip code 92011 there were 32 plans available for a certain cholesterol-lowering drug. Their annual plan costs (in dollars sorted from lowest to highest) were:

150	154	156	161	171	172	174	174
176	179	183	195	195	197	198	200
202	208	209	212	220	225	228	234
235	248	249	250	250	271	282	284

Because these are 3-digit numbers, we will use a *two-digit* stem. The stem is the first two digits of each price: 15, 16, ... , 28.

Frequency	Stem	Leaf
3	15	0 4 6
1	16	1
6	17	1 2 4 4 6 9
1	18	3
4	19	5 5 7 8
4	20	0 2 8 9
1	21	2
3	22	0 5 8
2	23	4 5
2	24	8 9
2	25	0 0
0	26	
1	27	1
2	28	2 4

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For example, the 3rd stem contains six prices: 171, 172, 174, 174, 176, 179. Besides requiring a two-digit stem, this display is long because the range is wide and there is not much central tendency. Another type of display (histogram or dot plot) might be easier to interpret, as well as being more compact.

A stem-and-leaf plot well for integer data with a limited range but becomes awkward for large samples, decimal data (e.g., \$6.39), or multi-digit data (e.g., \$3,857). In such cases, it is necessary to round the data to make the display “work.” For this reason, the stem-and-leaf plot is rarely seen in presentations of business data.

Example: Surgical Recovery Times

Hospital facility scheduling requires predicting post-anesthesia recovery times. Here is a sorted sample of actual post-anesthesia recovery time for 96 hospital patients who had surgery on a particular Friday:

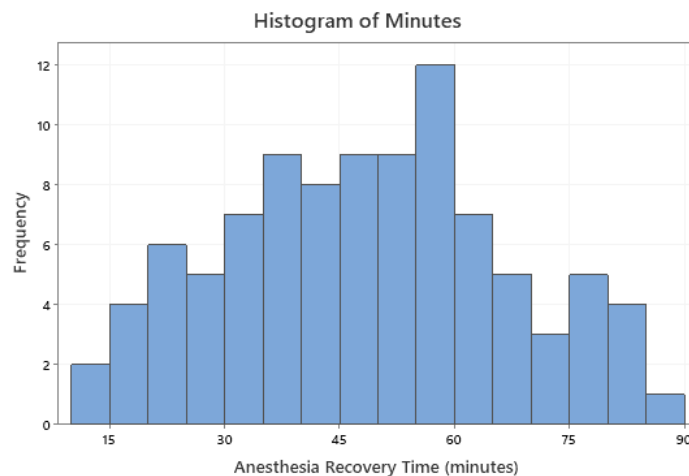
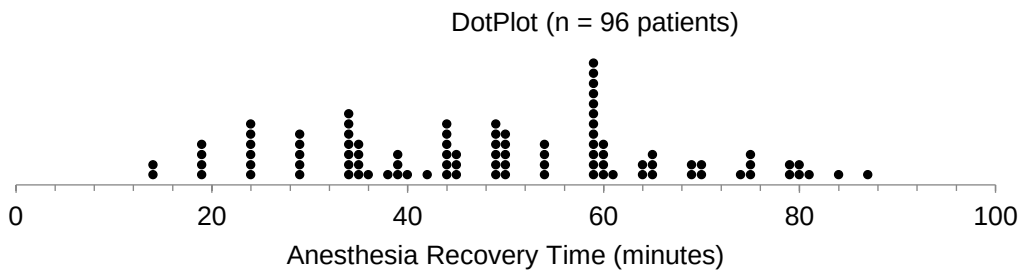
14	14	19	19	19	19	24	24	24	24	24	24
29	29	29	29	29	34	34	34	34	34	34	34
35	35	35	35	36	38	39	39	39	40	42	44
44	44	44	44	44	45	45	45	49	49	49	49
49	49	50	50	50	50	50	54	54	54	54	59
59	59	59	59	59	59	59	59	59	59	59	60
60	60	60	61	64	64	65	65	65	69	69	70
70	74	75	75	75	79	79	80	80	81	84	87

A hospital can use this type of information for facility scheduling, to be sure that enough beds and ward space will be available for surgical recovery.

The stem-and-leaf plot suggests a symmetric distribution. The modal class is 50-59 minutes (21 patients). A majority of patients recover in less than an hour (60 minutes). This sample is large enough to reveal the approximate shape of the population.

Frequency	Stem	Leaf
6	1	4 4 9 9 9 9
11	2	4 4 4 4 4 4 9 9 9 9 9
16	3	4 4 4 4 4 4 4 5 5 5 5 6 8 9 9 9
17	4	0 2 4 4 4 4 4 4 5 5 5 9 9 9 9 9 9
21	5	0 0 0 0 0 4 4 4 4 9 9 9 9 9 9 9 9 9 9 9
12	6	0 0 0 0 1 4 4 5 5 5 9 9
8	7	0 0 4 5 5 5 9 9
5	8	0 0 1 4 7
96		

For comparison, consider this dot plot (from MegaStat) and histogram (from Minitab). Which display do you feel would be most helpful for hospital surgical scheduling?



Do-It-Yourself Exercises

Exercise 1

Make a stem-and-leaf plot for the birth weights (ounces) of these 24 randomly chosen babies. Describe its appearance.

122 137 119 95 103 157 122 107 129 115 118 136
84 123 135 135 137 146 118 107 147 86 69 120

Exercise 2

Make a stem-and-leaf plot for these 29 wait times for IRS taxpayer assistance. Describe its appearance.

21 23 14 17 12 11 14 11 12 11 10 12 17 17 18
20 22 22 17 21 17 19 21 21 18 20 19 30 30

Hint: These are small data sets. Their stem-and-leaf displays can easily be done without a computer. To set up the stems, first locate the smallest and largest data value.

ANSWERS ON NEXT PAGE

Answers to Exercises

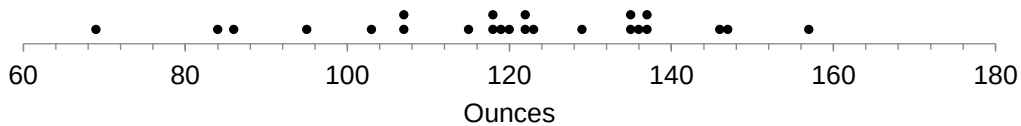
Exercise 1

Appearance is slightly left-skewed. The lower tail is longer and there is one unusually small birth weight. The dot plot reveals more details, but the stem-and-leaf is better at showing the shape of the sample because data values are in bins (like a histogram).

Frequency	Stem	Leaf
1	6	9
0	7	
2	8	4 6
1	9	5
3	10	3 7 7
4	11	5 8 8 9
5	12	0 2 2 3 9
5	13	5 5 6 7 7
2	14	6 7
1	15	7

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DotPlot



Exercise 2

There is possible positive skewness. Data values are concentrated on low end but there are a couple of high wait times. The dot plot reveals more detail.

Frequency	Stem	Leaf
18	1	0 1 1 1 2 2 2 4 4 7 7 7 7 7 8 8 9 9
9	2	0 0 1 1 1 1 2 2 3
2	3	0 0

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DotPlot

